

Resistors in series

When resistors are connected **in series**, each additional resistor effectively increases the length of the path taken by the charges, and so the more resistors you add, the greater the resistance becomes.

In other words,

$$R = \sum_1^n R_n$$

This relationship can be derived from K_1 and K_2 :

$$V = V_1 + V_2 + V_3 + \dots IR = (IR)_1 + (IR)_2 + (IR)_3 + \dots$$

And because I is the same (from K_1), we get

$$R = R_1 + R_2 + R_3 + \dots$$

Resistors in parallel

When resistors are connected **in parallel**, the total resistance drops because the additional resistor provides another path for the current.

By using the same approach we took earlier,

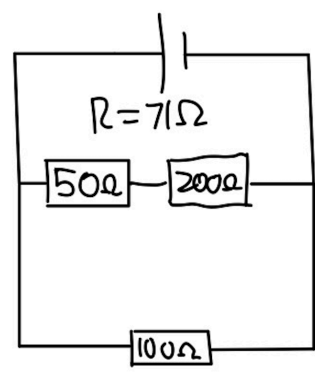
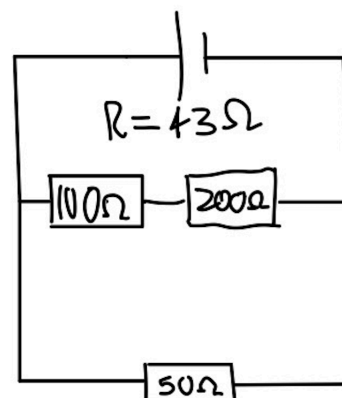
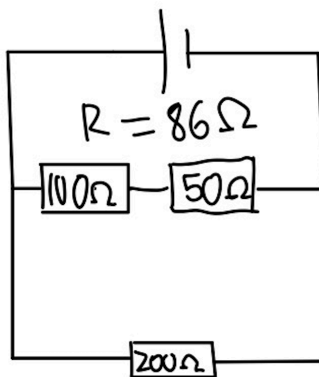
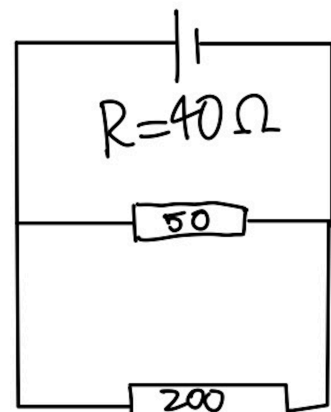
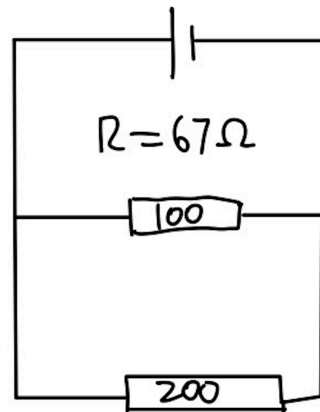
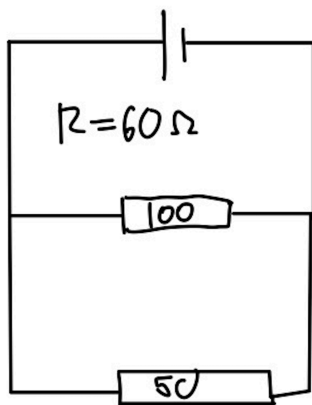
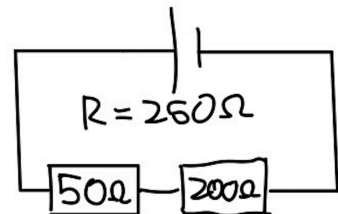
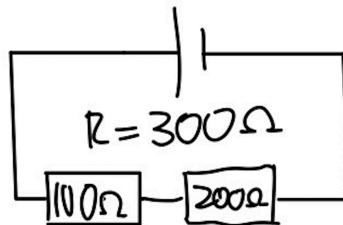
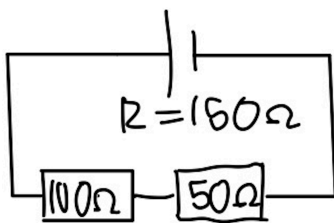
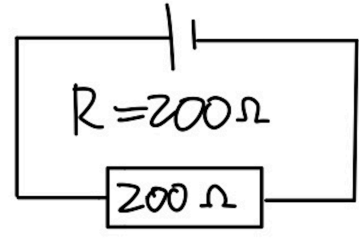
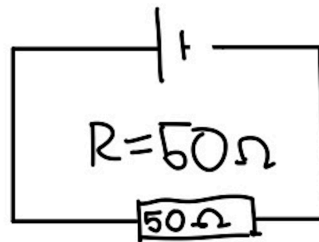
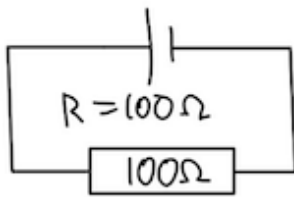
$$I = I_1 + I_2 + I_3 + \dots$$

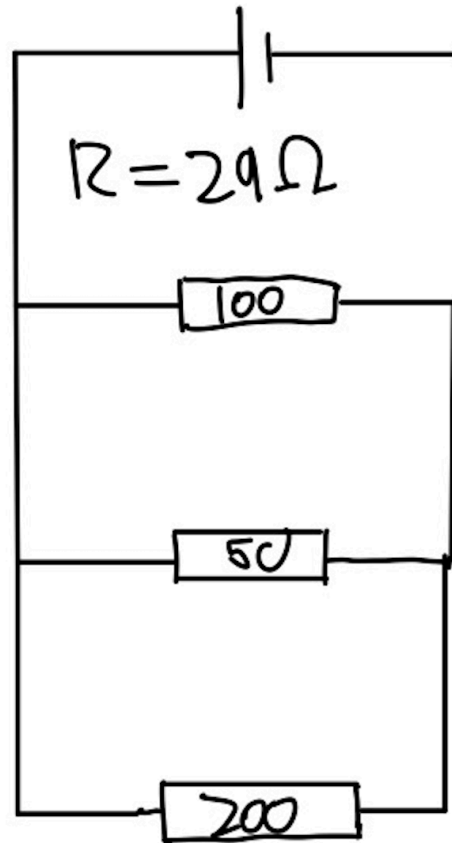
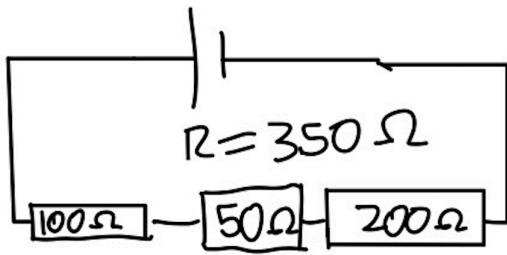
Since the resistors are in parallel, they have the same voltage and therefore

$$\frac{I}{V} = \frac{I_1}{V} + \frac{I_2}{V} + \frac{I_3}{V} + \dots \frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

Summary Questions

1. a. The resistance increases.
b. The resistance decreases.
2. a. $5 + 9 = 14\Omega$
b. $\left(\frac{1}{6} + \frac{1}{4}\right)^{-1} = 2.4\Omega$
c. $\left(\frac{1}{10} + \frac{1}{15} + \frac{1}{20}\right)^{-1} = 4.6\Omega$
3. a. $\left(\frac{1}{150+100} + \frac{1}{200}\right)^{-1} = 110\Omega$
b. $60 + \left(\frac{1}{20+10} + \frac{1}{20}\right)^{-1} = 72\Omega$
c. $\left(\frac{1}{10+10+10} + \frac{1}{10+10+10} + \frac{1}{10+10+10}\right)^{-1} = 10\Omega$
- 4.





$$1. \quad \frac{1}{2200} + \frac{1}{4700} + \frac{1}{R_3} = \frac{1}{1030}$$

$$\frac{1}{R_3} = \frac{1}{1030} - \left(\frac{1}{2200} + \frac{1}{4700} \right)$$

$$\frac{1}{R_3} = 0.0003035623744$$

$$R_3 = 3300\Omega$$